Chico’s Barbershop

XISD6329

Work Integrated Learning 3B

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1. Introduction

This report describes the implementation of Chico's Barbershop, an Android application supported by a companion website. The project builds on the requirements and design work completed in Task 1. The development work described here covers the user interface, account management, appointment booking, database design, testing and DevOps.

The project materials identify Kotlin, Android Studio, Android SDK 35, Gradle 8.14.4, Firebase, GitHub and Azure DevOps as the main technologies. The final report should be supported by evidence from the current repository, a successful build, the live Firebase project and a physical-device demonstration.

2. Team Meetings and Peer Evaluation

The team used physical and virtual meetings to coordinate development, review progress and assign responsibilities. Azure DevOps was used to track work, while GitHub supported source-code management. Replace the phase labels below with actual meeting dates and attach supporting records where available.

Meeting / Phase	Format	Main activities
Kick-off	Campus meeting	Reviewed Task 1 feedback, created the GitHub repository and set up the Azure DevOps board.

Database and authentication	Online meeting	Reviewed Firebase setup, data structure, registration, login and Google Sign-In.
Core modules and interface	Campus meeting	Worked on screens, input validation, booking features, settings and website updates.
CI/CD and device testing	Online / campus	Reviewed automated builds and tests, installed the app on a physical phone and planned the demonstration.

2.1 Task Allocation and Peer Evaluation

Team member Name	Responsibility	Peer score (/10)
Liam Frans - Developer	Registration, login, password protection and Firebase Authentication.	Complete 10/10
Liam Frans - Developer	GUI, input validation, crash prevention and user settings.	Complete 10/10
Babalo Jama – Developer	Database structure, sample data, API and appointment booking.	Complete 10/10
Babalo Jama – Developer	GitHub Actions, unit tests, Help/FAQ and website updates.	Complete 10/10

Figure 1: Team Members Roles

3. System Implementation

The application is described as a Kotlin project developed in Android Studio, with XML-based layouts and Material Design components. The interface should make the main tasks easy to find and use on an Android phone.

3.1 Graphical User Interface

- **Consistent design:** Use barbershop branding, a consistent colour palette and readable text.
- **Clear navigation:** Provide straightforward access to the home screen, services, booking calendar, settings and Help section.
- **Responsive layout:** Check that controls remain readable and usable on different physical screen sizes.

3.2 Input Validation and Error Handling

Input validation helps prevent invalid data and reduces the risk of application crashes. The implementation described in the source includes:

- **Required fields:** Check that mandatory fields are not blank before processing a form.
- **Email validation:** Validate email addresses before registration or login.
- **Phone numbers:** Use an appropriate phone input type and validate the required number format.
- **Booking dates:** Prevent users from selecting dates in the past.
- **Network errors:** Handle Firebase or network failures and show a clear message instead of allowing an unhandled exception.

3.3 Main Application Modules

Module	Purpose
Authentication	Supports registration, email/password login, password reset and Google Sign-In where configured.
Services and barbers	Displays available services, prices and barber profiles from the database.
Appointment booking	Allows a customer to select a service, barber, date and available time slot, then confirm or cancel a booking.
User settings	Allows an authenticated user to update profile details and manage available preferences.

4. Security: Authentication and Password Protection

The report identifies Firebase Authentication as the account-management service. Passwords should not be stored as plain text by the application. Firebase Authentication manages password credentials on its authentication service, while secure network connections protect data in transit.

Google Sign-In is described as an additional sign-in option. Before submission, verify that the feature is enabled in the Firebase project, that the required Android configuration is present, and that both successful and unsuccessful sign-in attempts have been tested.

Avoid describing passwords as simply "encrypted" if the implementation uses password hashing. In the report, use "password hashing and secure credential management" unless the exact mechanism has been verified from the service documentation.

5. Database Design and Sample Data

The source describes Firebase as the online data service and lists four main data groups: Users, Barbers, Services and Bookings. The precise Firebase product used (Cloud Firestore or Realtime Database) must be stated correctly in the final submission.

Collection / table	Main fields described
Users Login	userId, fullName, email, phone, authProvider, notificationsEnabled, preferredBarber
Bookings	userID, date, timeslot, price, rating, appointmentID, status, durationMins, createdAt
Customer	customerNamec, ustomerEmail, phoneNumber

5.1 Users Log

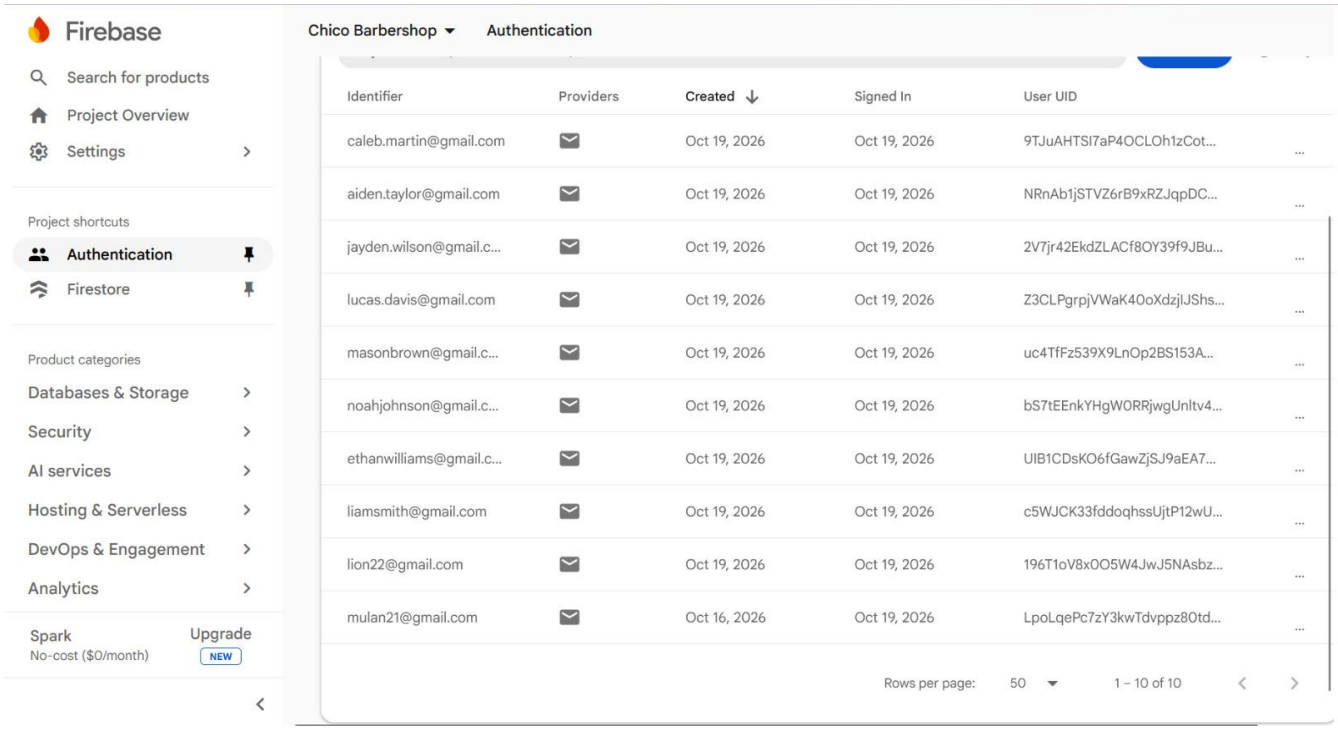


Figure 2: List of Users on Database upon Login

Google Single sign-on fully fuctional. Firebase Authentication manages password credentials on its authentication service, while secure network connections protect data in transit.

5.2 Sample Bookings

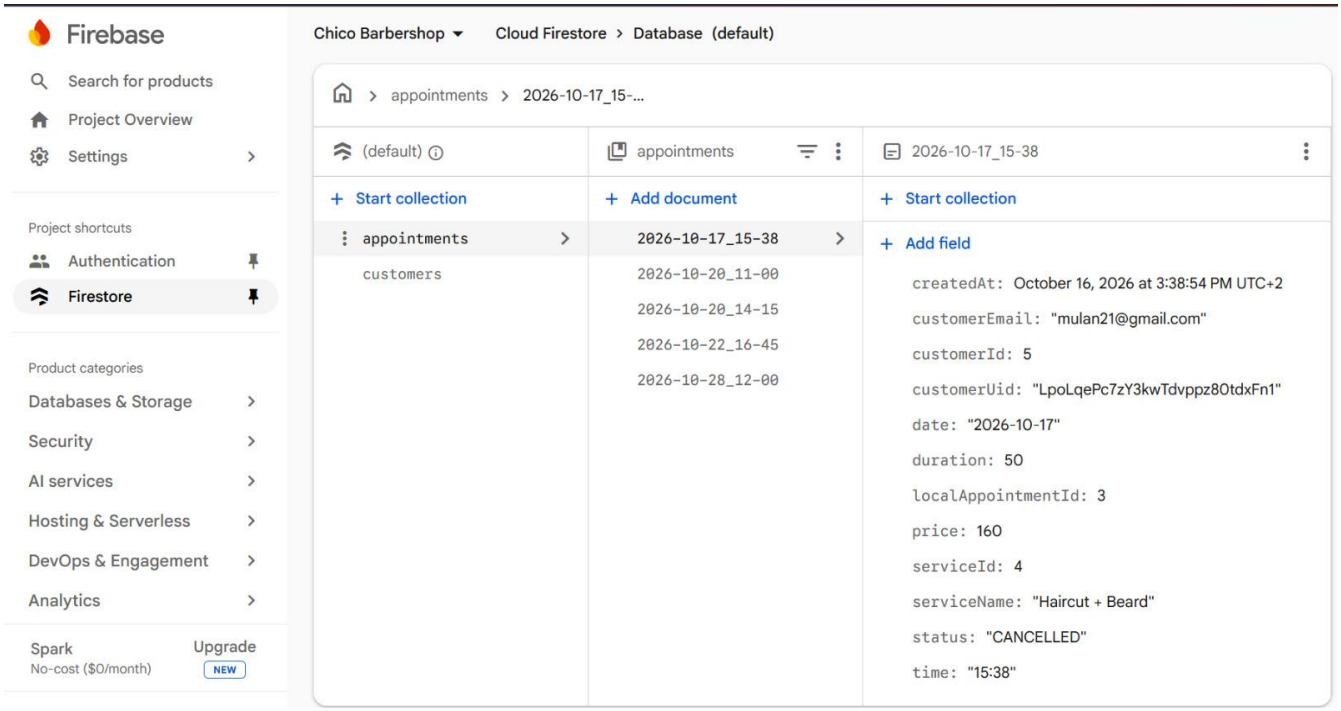
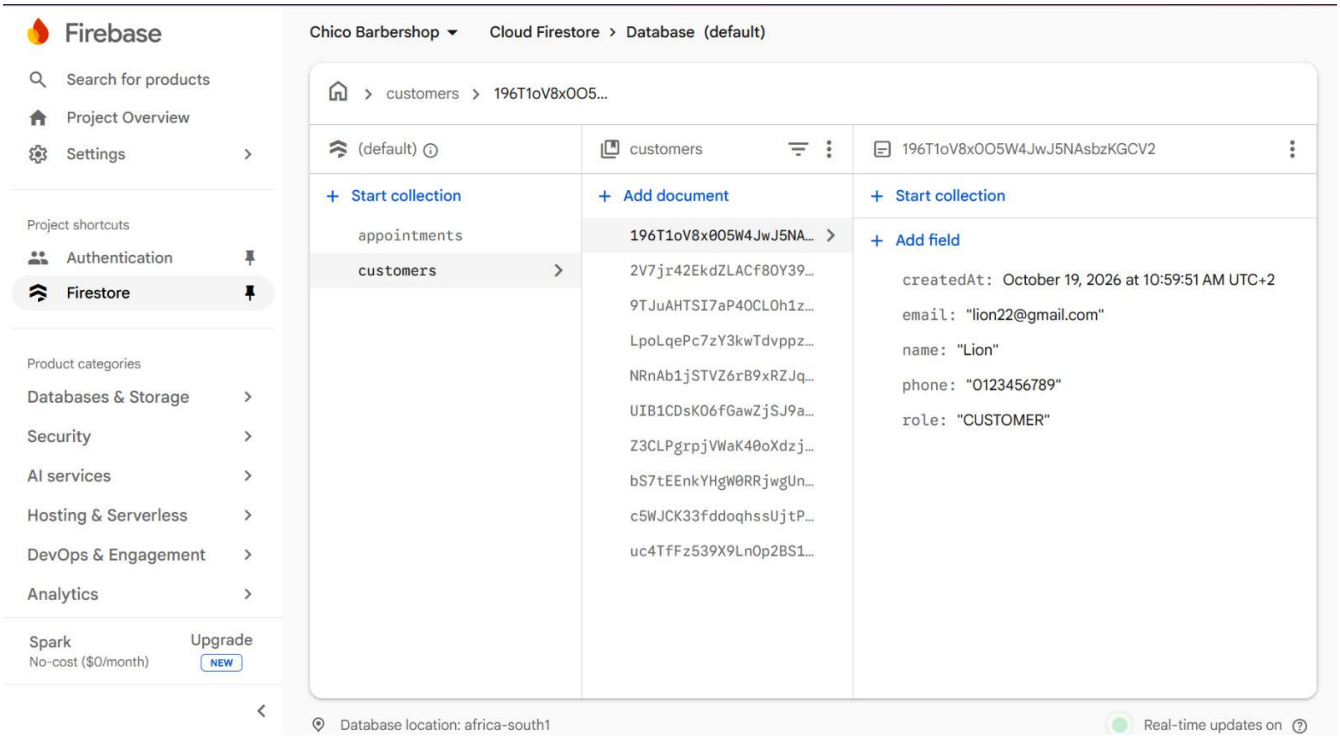


Figure 3: Appointment/ Bookings Log

5.3 Customer



6. Testing and Crash Prevention

Testing should confirm that the app handles valid and invalid inputs, authentication failures, booking conflicts and settings updates. The following cases are taken from the supplied report and should be marked as passed only after they have been executed.

Test	Input / condition	Expected result	Evidence / result
Registration validation	Blank email or invalid/short password	Registration is rejected and a clear validation message appears.	Record actual result
Valid registration	Valid user details and password meeting configured rules	Account is created successfully; credentials are managed by Firebase Authentication.	Record actual result
Login / Google Sign-In	Valid credentials or valid Google sign-in flow	User is authenticated and directed to the appropriate screen.	Record actual result
Booking conflict	Select a time slot that is already taken	The app prevents a duplicate booking and asks the user to choose another slot.	Record actual result
Settings update	Change profile details or notification preference	The change is saved and displayed correctly after reopening the screen.	Record actual result
Network failure	Disable connectivity during a data request	The app displays a useful error message and remains usable.	Record actual result

6.1 Workflow Testing

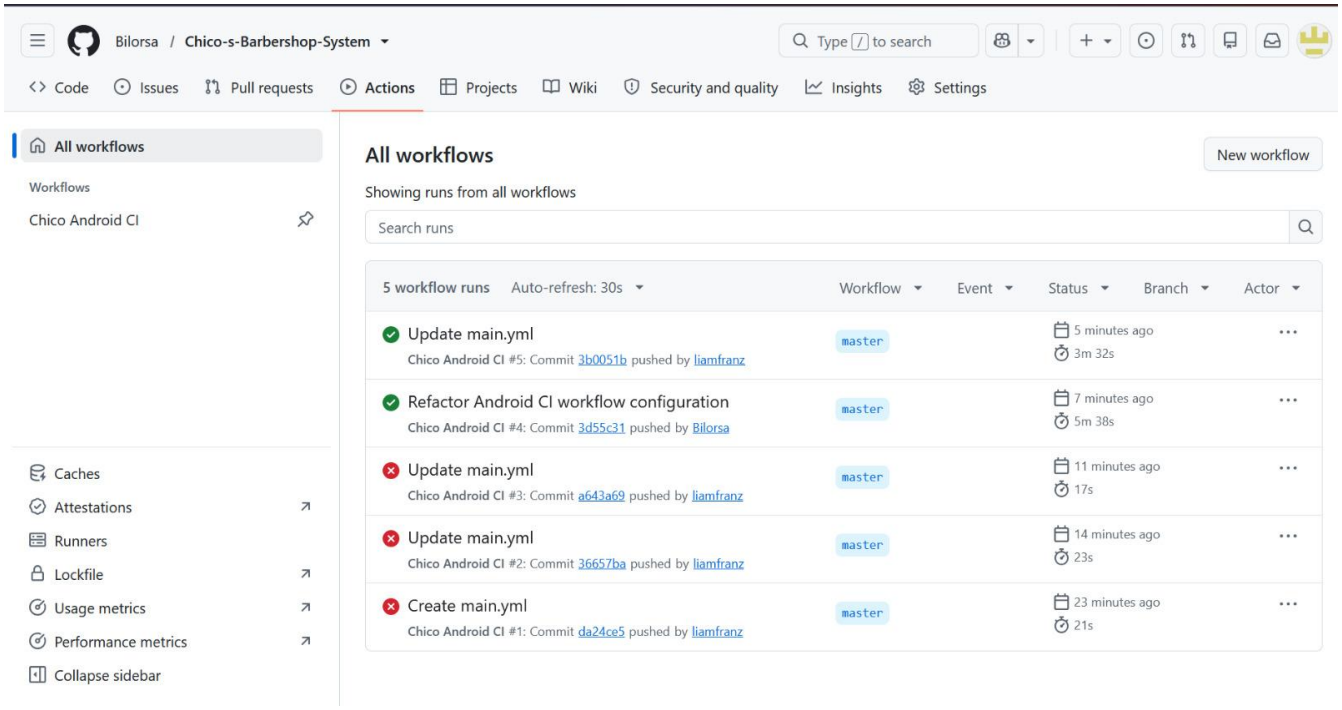


Figure 4: Test Build

The project describes a DevOps workflow that connects planning, source control, building, testing, release and deployment. Azure DevOps is used for planning and tracking, while GitHub stores the source code and hosts the proposed CI workflow.

7. User Support and User Guide

7.1 Help and Support

The planned Help section provides answers to common questions, including how to book an appointment, reset a password, and cancel or reschedule a booking. If implemented, the screen should also provide a working contact option for Chico's Barbershop.

7.2 Basic User Instructions

1. **Create an account or sign in:** Open the application and register with the required details, or use Google Sign-In if available.
2. **Book an appointment:** Choose a service, select a barber, choose an available date and time, then confirm the booking.
3. **Manage account settings:** Open Settings to update the profile details or preferences supported by the application.
4. **Get help:** Open the Help section and follow the relevant instructions. Use the contact option if further assistance is needed.

Conclusion

The Chico Barbershop Application demonstrates how an Android mobile application can support the digital management of a barbershop. By providing customer and administrator interfaces, the system aims to make appointment-related activities more convenient and administrative functions easier to access.

The project combines mobile application development with account authentication, database operations, interface design, and software testing. Its success depends on more than the appearance of its screens: the underlying processes must correctly validate user input, identify customers, store records, manage appointments, and protect access to administrative functions.

Testing is therefore an essential part of the project. Unit tests, integration tests, functional tests, database tests, and manual device testing provide different forms of evidence about the application's quality. Automated workflows can further improve consistency by running configured tests and build checks whenever code changes.

Future improvements could focus on stronger appointment conflict prevention, reliable customer identification, working notifications, enhanced reporting, and broader automated test coverage. These improvements would help make Chico a more reliable and maintainable solution for managing barbershop activities.

Supporting Links

Repository: Chico-s-Barbershop-System-master (GitHub)

Video Explaining Application: https://youtu.be/VoznU7Iesu8